

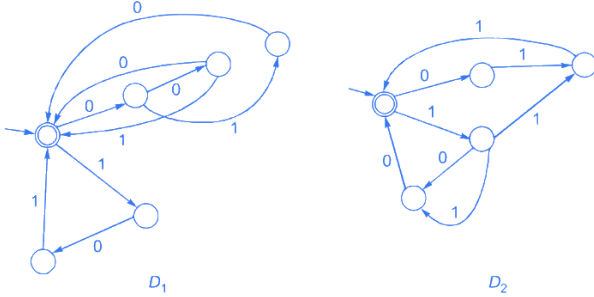
FINITE AUTOMATA AND REGULAR LANGUAGE

(2 Marks)

1. GATE CSE 2026 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the following two finite automata D_1 and D_2 .



Which of the following statements is/are true?

- (A) $\angle(D_1) = \angle(D_2)$
- (B) $\angle(D_1)$ is a proper subset of $\angle(D_2)$
- (C) $\angle(D_1) \cap L(D_2) = \{\epsilon\}$
- (D) $(L(D_1) \cup L(D_2))^*$ consists of all strings in $\{0, 1\}^*$ whose length is divisible by 3.

2. GATE CSE 2026 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0

Let L_1 and L_2 be two languages over a finite alphabet, such that $L_1 \cap L_2$ and L_2 are regular languages.

Which of the following statements is/are always true?

- (A) L_1 is regular
- (B) $L_1 \cup L_2$ is regular
- (C) $\overline{L_2}$ is context free
- (D) L_1 is context free

3. GATE CSE 2025 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#_\alpha(x)$ denote the number of occurrences of α in x . Which one or more of the following option(s) define(s) regular language(s)?

- (A) $\{a^m b^n \mid m, n \geq 0\}$
- (B) $\{a, b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
- (C) $\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2(\text{mod } 7), \text{ and } \#_b(w) \equiv 3(\text{mod } 9)\}$
- (D) $\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2(\text{mod } 7), \text{ and } \#_b(w) = \#_a(w)\}$

4. GATE CSE 2025 Set 2

Numerical Marks: +2 / -0

Let $\Sigma = \{1, 2, 3, 4\}$. For $x \in \Sigma^*$, let $\text{prod}(x)$ be the product of symbols in x modulo 7. We take $\text{prod}(\epsilon) = 1$, where ϵ is the null string.

For example, $\text{prod}(124) = (1 \times 2 \times 4) \text{ mod } 7 = 1$.

Define $L = \{x \in \Sigma^* \mid \text{prod}(x) = 2\}$.

The number of states in a minimum state DFA for L is _____ (Answer in integer)

5. GATE CSE 2025 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0

Consider the following two languages over the alphabet $\{a, b\}$:

$$L_1 = \{\alpha\beta\alpha \mid \alpha \in \{a, b\}^+ \text{ AND } \beta \in \{a, b\}^+\}$$

$$L_2 = \{\alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a, b\}^+\}$$

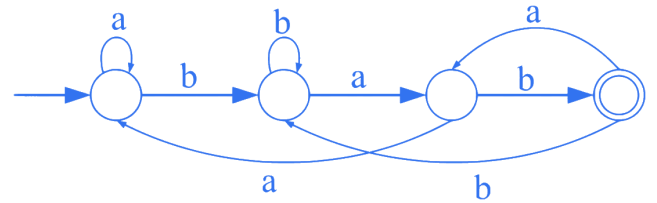
Which ONE of the following statements is CORRECT?

- (A) Both L_1 and L_2 are regular languages.
- (B) L_1 is a regular language but L_2 is not a regular language.
- (C) L_1 is not a regular language but L_2 is a regular language.
- (D) Neither L_1 nor L_2 is a regular language.

6. GATE CSE 2025 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.



- (A) The set of all strings containing an even number of b 's.
- (B) The set of all strings containing the pattern bab .
- (C) The set of all strings ending with the pattern bab .
- (D) The set of all strings not containing the pattern aba .

7. GATE CSE 2025 Set 1

Numerical Marks: +2 / -0

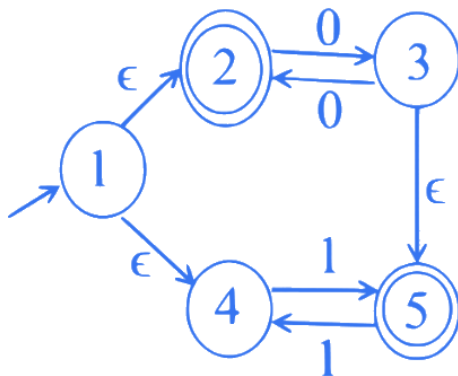
Consider a finite state machine (FSM) with one input X and one output f , represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer)

Present state	Next state		Output f	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

8. GATE CSE 2024 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.66

Let M be the 5-state NFA with ϵ -transitions shown in the diagram below.



Which one of the following regular expressions represents the language accepted by M ?

- (A) $(00)^* + 1(11)^*$
- (B) $0^* + (1 + 0(00)^*)(11)^*$
- (C) $(00)^* + (1 + (00)^*)(11)^*$
- (D) $0 + 1(11)^* + 0(11)^*$

9. GATE CSE 2024 Set 2

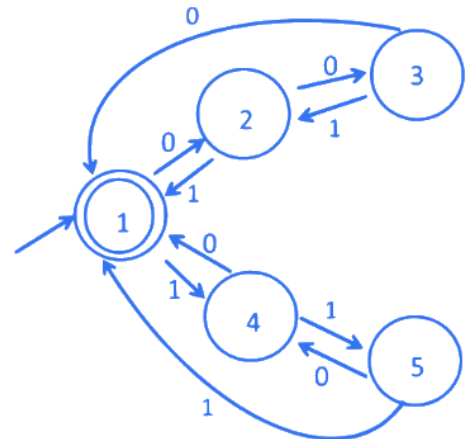
Numerical Marks: +2 / -0

Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{w \in (a+b)^* \mid |w| \leq 4\}$, where $|w|$ denotes the length of string w . The number of strings in L_2 which are also in L_1 is _____.

10. GATE CSE 2024 Set 1

MCQ (More than One Correct Answer) Marks: +2 / -0

Consider the 5-state DFA M accepting the language $L(M) \subseteq (0+1)^*$ shown below. For any string $w \in (0+1)^*$ let $n_0(w)$ be the number of 0's in w and $n_1(w)$ be the number of 1's in w .



Which of the following statements is/are FALSE?

- (A) States 2 and 4 are distinguishable in M
- (B) States 3 and 4 are distinguishable in M
- (C) States 2 and 5 are distinguishable in M
- (D) Any string w with $n_0(w) = n_1(w)$ is in $L(M)$

11. GATE CSE 2024 Set 1

Numerical Marks: +2 / -0

Consider the following two regular expressions over the alphabet $\{0,1\}$:

$$r = 0^* + 1^*$$

$$s = 01^* + 10^*$$

The total number of strings of length less than or equal to 5, which are neither in r nor in s , is _____.

12. GATE CSE 2023

Numerical Marks: +2 / -0

Consider the language L over the alphabet $\{0, 1\}$, given below:

$$L = \{w \in \{0, 1\}^* \mid w \text{ does not contain three or more consecutive } 1's\}.$$

The minimum number of states in a Deterministic Finite-State Automaton (DFA) for L is _____.

13. GATE CSE 2021 Set 2

MCQ (More than One Correct Answer) Marks: +2 / -0

Which of the following regular expressions represent(s) the set of all binary numbers that are divisible by three? Assume that the string ϵ is divisible by three.

- (A) $(0 + 1(01^*0)^*1)^*$
- (B) $(0^*(1(01^*0)^*1)^*)^*$
- (C) $(0 + 11 + 10(1 + 00)^*01)^*$
- (D) $(0 + 11 + 11(1 + 00)^*00)^*$

14. GATE CSE 2021 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.66

Consider the following two statements about regular languages:

S_1 : Every infinite regular language contains an undecidable language as a subset.

S_2 : Every finite language is regular.

Which one of the following choices is correct?

- (A) Only S_2 is true.
- (B) Neither S_1 nor S_2 is true.
- (C) Only S_1 is true.
- (D) Both S_1 and S_2 are true.

15. GATE CSE 2021 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0.67

Consider the following language.

$L = \{ w \in \{0, 1\}^* \mid w \text{ ends with the substring } 011 \}$

Which one of the following deterministic finite automata accepts L ?

- (A)
- (B)
- (C)
- (D)

16. GATE CSE 2020

Numerical Marks: +2 / -0

Consider the following language.

$L = \{x \in \{a, b\}^* \mid \text{number of } a\text{'s in } x \text{ is divisible by 2 but not divisible by 3}\}$

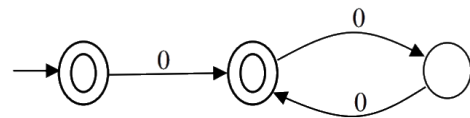
The minimum number of states in a DFA that accepts L is _____.

17. GATE CSE 2018

Numerical Marks: +2 / -0

Given a language L , define L^i as follows: $L^0 = \{\epsilon\}$
 $L^i = L^{i-1} \cdot L$ for all $i > 0$

The order of a language L is defined as the smallest k such that $L^k = L^{k+1}$. Consider the language L_1 (over alphabet 0) accepted by the following automaton.



The order of L_1 is _____.

18. GATE CSE 2016 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following two statements :

I. If all states of an NFA are accepting states then the language accepted by the NFA is Σ^* .

II. There exists a regular language A such that for all languages B , $A \cap B$ is regular.

Which one of the following is **CORRECT**?

- (A) Only I is true
- (B) Only II is true
- (C) Both I and II are true
- (D) Both I and II are false

19. GATE CSE 2015 Set 1

Numerical Marks: +2 / -0



Consider the DFAs M and N given above. The number of states in a minimal DFA that accepts the language $L(M) \cap L(N)$ is _____.

20. GATE CSE 2015 Set 2

Numerical Marks: +2 / -0

The number of states in the minimal deterministic finite automaton corresponding to the regular expression $(0 + 1)^* (10)$ is _____.

21. GATE CSE 2015 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the alphabet $\Sigma = \{0, 1\}$, the null/empty string λ and the sets of strings X_0 , X_1 , and X_2 generated by the corresponding non-terminals of a regular grammar. X_0 , X_1 , and X_2 are related as follows.

$$\begin{aligned} X_0 &= 1 X_1 \\ X_1 &= 0 X_1 + 1 X_2 \\ X_2 &= 0 X_1 + \{\lambda\} \end{aligned}$$

Which one of the following choices precisely represents the strings in X_0 ?

- (A) $10(0^* + (10)^*)1$
- (B) $10(0^* + (10)^*)^*1$
- (C) $1(0 + 10)^*1$
- (D) $10(0 + 10)^*1 + 110(0 + 10)^*1$

22. GATE CSE 2015 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

Which of the following languages is/are regular?
 $L_1 : \{wxw^R \mid w, x \in \{a, b\}^* \text{ and } |w|, |x| > 0\}$, w^R is the reverse of string w

$$L_2 : \{a^n b^m \mid m \neq n \text{ and } m, n \geq 0\}$$

$$L_3 : \{a^p b^q c^r \mid p, q, r \geq 0\}$$

- (A) L_1 and L_3 only
- (B) L_2 only
- (C) L_2 and L_3 only
- (D) L_3 only

23. GATE CSE 2014 Set 2

MCQ (Single Correct Answer) Marks: +2 / -0.6

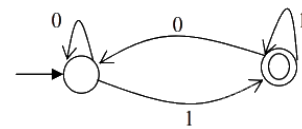
Let $L_1 = \{w \in \{0, 1\}^* \mid w \text{ has at least as many occurrences of } (110)'s \text{ as } (011)'s\}$. Let $L_2 = \{w \in \{0, 1\}^* \mid w \text{ has at least as many occurrences of } (000)'s \text{ as } (111)'s\}$. Which one of the following is TRUE?

- (A) L_1 is regular but not L_2
- (B) L_2 is regular but not L_1
- (C) Both L_1 and L_2 are regular
- (D) Neither L_1 nor L_2 are regular

24. GATE CSE 2014 Set 1

MCQ (Single Correct Answer) Marks: +2 / -0.6

Which of the regular expression given below represent the following DFA?



- I) $0^*1(1 + 00^*1)^*$
- II) $0^*1^*1 + 11^*0^*1$
- III) $(0 + 1)^*1$
- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

25. GATE CSE 2013

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following languages

$$L_1 = \{0^p 1^q 0^r \mid p, q, r \geq 0\}$$

$$L_2 = \{0^p 1^q 0^r \mid p, q, r \geq 0, p \neq r\}$$

Which one of the following statements is FALSE?

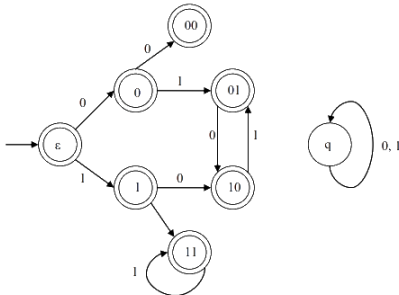
- (A) L_2 is context-free
- (B) $L_1 \cap L_2$ is context-free
- (C) Complement of L_2 is recursive
- (D) Complement of L_1 is context-free but not regular

26. GATE CSE 2012

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the set of strings on $\{0, 1\}$ in which, every substring of 3 symbols has at most two zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed *DFA* that accepts this language is shown below.

The missing arcs in the *DFA* are



(A)

	00	01	10	11	q
00	1	0			
01				1	
10	0				
11			0		

(B)

	00	01	10	11	q
00		0			1
01		1			
10				0	
11		0			

(C)

	00	01	10	11	q
00		1			0
01		1			
10			0		
11		0			

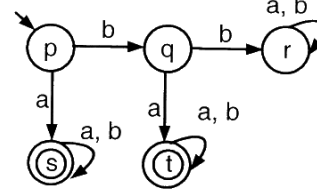
(D)

	00	01	10	11	q
00		1			0
01				1	
10	0				
11			0		

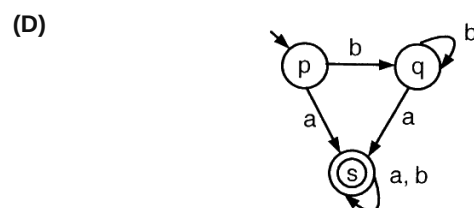
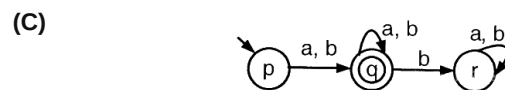
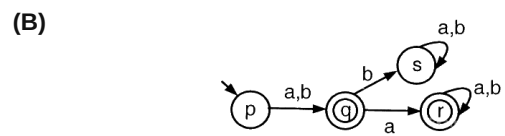
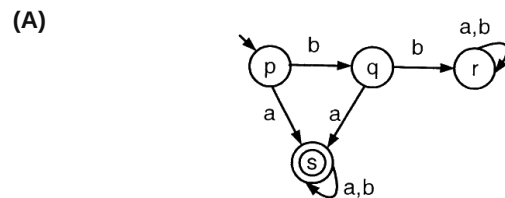
27. GATE CSE 2011

MCQ (Single Correct Answer) Marks: +2 / -0.6

A deterministic finite automaton (*DFA*) *D* with alphabet $\Sigma = \{a, b\}$ is given below



Which of the following finite state machines is a valid minimal *DFA* which accepts the same languages as *D*?



28. GATE CSE 2011

MCQ (Single Correct Answer) Marks: +2 / -0.6

Definition of the language *L* with alphabet $\{a\}$ is given as following. $L = \{a^{nk} \mid k > 0, n \text{ is a positive integer constant}\}$

What is the minimum number of states needed in a *DFA* to recognize *L*?

- (A) $k + 1$
- (B) $n + 1$
- (C) 2^{n+1}
- (D) 2^{k+1}

29. GATE CSE 2010

MCQ (Single Correct Answer) Marks: +2 / -0.6

Let w be any string of length n in $\{0, 1\}^*$. Let L be the set of all substrings of w . What is the minimum number of states in a non-deterministic finite automation that accepts L ?

- (A) $n - 1$
 (B) n
 (C) $n + 1$
 (D) 2^{n+1}

30. GATE CSE 2010

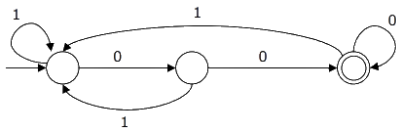
MCQ (Single Correct Answer) Marks: +2 / -0.6

Let $L = \{w \in (0 + 1)^* \mid w \text{ has even number of } 1's\}$, i.e L is the set of all bit strings with even number of $1's$. which one of the regular expression below represents L .

- (A) $(0^*10^*1)^*$
 (B) $0^*(10^*10^*)^*$
 (C) $0^*(10^*1)^*0^*$
 (D) $0^*1(10^*1)^*10^*$

31. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +2 / -0.6



The above *DFA* accepts the set of all strings over $\{0, 1\}$ that

- (A) Begin either with 0 or 1
 (B) End with 0
 (C) End with 00.
 (D) Contain the substring 00.

32. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +2 / -0.6

$L = L_1 \cap L_2$ where L_1 and L_2 are languages defined as follows.

$$L_1 = \{a^m b^m c a^n b^n \mid m, n \geq 0\}$$

$$L_2 = \{a^i b^j c^k \mid i, j, k \geq 0\}$$

- Then L is
- (A) Not recursive
 (B) Regular
 (C) Context free but not regular
 (D) Recursively enumerable but not context free

33. GATE CSE 2008

MCQ (Single Correct Answer) Marks: +2 / -0.6

Which of the following are regular sets?

- I. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\}$
 II. $\{a^n b^m \mid n = 2m\}$
 III. $\{a^n b^m \mid n \neq m\}$
 IV. $\{x c y \mid x, y \in \{a, b\}^*\}$

- (A) I & IV only
 (B) I & III only
 (C) I only
 (D) IV only

34. GATE CSE 2008

MCQ (Single Correct Answer) Marks: +2 / -0.6

Given below are two finite state automata ($\rightarrow$ indicates the start state and F indicates a final state)

Y:

	a	b
$\rightarrow 1$	1	2
2 (F)	2	1

Z:

	a	b
$\rightarrow 1$	2	2
2 (F)	1	1

Which of the following represents the product automaton ZY ?

(A)

	a	b
$\rightarrow P$	S	R
Q	R	S
R (F)	Q	P
S	Q	P

(B)

	a	b
$\rightarrow P$	S	Q
Q	R	S
R (F)	Q	P
S	P	Q

(C)

	a	b
$\rightarrow P$	Q	S
Q	R	S
R (F)	Q	P
S	Q	P

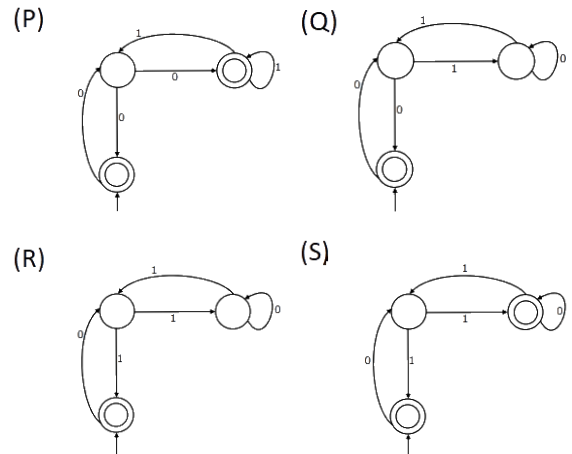
(D)

	a	b
$\rightarrow P$	S	Q
Q	S	R
R (F)	Q	P
S	Q	P

35. GATE CSE 2008

MCQ (Single Correct Answer) Marks: +2 / -0.6

Match the following NFA s with the regular expressions they correspond to



1. $\varepsilon + 0(01^*1 + 00)^*01^*$ 2. $\varepsilon + 0(10^*1 + 00)^*0$

3. $\varepsilon + 0(10^*1 + 10)^*1$ 4. $\varepsilon + 0(10^*1 + 10)^*10^*$

(A) $P - 2, Q - 1, R - 3, S - 4$

(B) $P - 1, Q - 3, R - 2, S - 4$

(C) $P - 1, Q - 2, R - 3, S - 4$

(D) $P - 3, Q - 2, R - 1, S - 4$

36. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

Which of the following languages is regular?

(A) $\{ww^R \mid w \in \{0, 1\}^+\}$

(B) $\{ww^Rx \mid x, w \in \{0, 1\}^+\}$

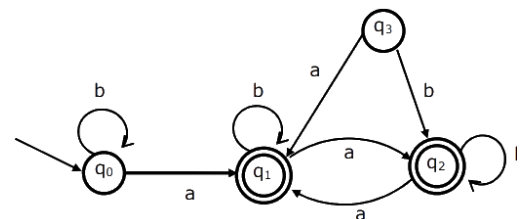
(C) $\{xww^R \mid x, w \in \{0, 1\}^+\}$

(D) $\{xww^R \mid x, w \in \{0, 1\}^+\}$

37. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following finite state automation



The minimum state automation equivalent to the above FSA has the following number of states

(A) 1

(B) 2

(C) 3

(D) 4

38. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

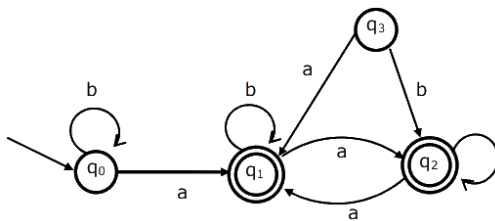
A minimum state deterministic finite automation accepting the language $L = \{w \mid w \in \{0, 1\}^*, \text{ number of } 0\text{'s and } 1\text{'s in } w \text{ are divisible by } 3 \text{ and } 5, \text{ respectively} \}$ has

- (A) 15 states
 - (B) 11 states
 - (C) 10 states
 - (D) 9 states
-

39. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +2 / -0.6

Consider the following finite state automation



The language accepted by this automation is given by the regular expression

- (A) $b^*ab^*ab^*ab^*$
 - (B) $(a + b)^*$
 - (C) $b^*a(a + b)^*$
 - (D) $b^*ab^*ab^*$
-